

MATH 151: ALGEBRA

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Algebra of Complex Numbers

We find that the set of Reals, **R** is not closed under the extraction of some roots. In particular, the problem:

$$x^2 = -1 \quad (1)$$

has no solution in **R**.

Hence we enlarge **R**, so that the square roots of negative numbers can be extracted. Thus we look for an enlarged set, **C**, called the Complex numbers.



Cont...

Complex numbers arise most directly out of solving polynomial equations.

Consider the quadratic equation

$$z^2 - 4z + 5 = 0 \quad (2)$$



Cont...

The equation above has two solutions z_1 and z_2 such that

$$(z - z_1)(z - z_2) = 0 \quad (3)$$

Using the familiar formula for the roots of a quadratic equation , the solutions z_1 and z_2 written in brief as

$$z_{1,2} = \frac{4 \pm \sqrt{(-4)^2 - 4(1 \times 5)}}{2} = 2 \pm \frac{\sqrt{-4}}{2} \quad (4)$$



Cont...

The second term on the RHS of (4) is called an **imaginary term** since it contains the square root of a negative number, the first term is called a **real term**. The full solution is the sum of a real term and an imaginary term and it is called a **Complex Number**.

Notation

The choice of the symbol z for the quadratic variable was not arbitrary; the conventional representation of a complex number is z , where z is the sum of a real part x and i times an imaginary part y , i.e.

$$z = x + iy, \quad (5)$$

where i is used to denote the square root of -1 .

Cont...

The real part x and the imaginary part y are usually denoted by $\text{Re}(z)$ and $\text{Im}(z)$ respectively.

Engineers, in particular, use j instead of i . However, for consistency, we will use i .

Cont...

In our particular example $\sqrt{-4} = 2\sqrt{-1} = 2i$,
and hence the two solutions of (3) are

$$z_{1,2} = 2 \pm \frac{2i}{2} = 2 \pm i \quad (6)$$

Thus, here $x = 2$ and $y = \pm 1$.



Cont...

For compactness, a complex number is sometimes written in the form

$$z = (x, y), \quad (7)$$

where the components z may be thought of as coordinates in an xy -plot.



Addition And Subtraction of Complex Numbers

- The addition of two complex numbers z_1 and z_2 , in general, gives another complex number.
- The real and imaginary parts add similarly to real numbers:

$$z_1 + z_2 = (x_1 + y_1 i) + (x_2 + y_2 i) = (x_1 + x_2) + (y_1 + y_2)i \quad (8)$$



Cont...

Or in component notation:

$$z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad (9)$$

By straightforward application of commutativity and associativity, i.e.,

$$\begin{aligned} z_1 + z_2 &= z_2 + z_1 \\ z_1 + (z_2 + z_3) &= (z_1 + z_2) + z_3 \end{aligned}$$



Example 1

Question:

Sum the complex numbers $1 + 2i$, $3 - 4i$, $-2 + i$.

Solution:

Summing the real terms: $1 + 3 - 2 = 2$,

Summing the imaginary terms: $2i - 4i + i = -i$,

Hence, $(1 + 2i) + (3 - 4i) + (-2 + i) = 2 - i$.

The subtraction of complex numbers is very similar to their addition.

Equality

Let $z_1 = x_1 + y_1i$ and $z_2 = x_2 + y_2i$ be complex numbers, then:

$$z_1 = z_2$$

if and only if

$$x_1 = x_2 \quad \text{and} \quad y_1 = y_2$$



Modulus And Argument

The **modulus** of the complex number z is denoted by $|z|$ and is defined as

$$|z| = \sqrt{x^2 + y^2}. \quad (10)$$

Hence the modulus of the complex number is the distance of the corresponding point from the origin in the Argand diagram.

Cont...

The **argument** of the complex number z is denated by **arg z** and is defined as

$$\text{arg}z = \tan^{-1} \left(\frac{y}{x} \right) \quad (11)$$

It can be seen that $\arg z$ is the angle that the line joining the origin to z on the Argan diagram makes with the positive $x - axis$. The anticlockwise direction is taken to be positive by convention.

Cont...

Account must be taken of the signs of x and y individually in determining in which quadrant $\arg z$ lies.

Example 2

Question

Find the modulus and the argument of the complex number $z = 2 - 3i$.

Cont...

Solution

The modulus is given by

$$|z| = \sqrt{(2)^2 + (-3)^2} = \sqrt{13}$$

the argument is given by

$$\arg z = \theta = \tan^{-1} \left(\frac{-3}{2} \right) = -56.31$$



Cont...

Since $x = 2$ and $y = -3$, z clearly lies in the fourth quadrant; (i.e., the argument must be measured from the positive real axis), therefore $\arg z = 56.31^\circ$ is the appropriate answer.



Multiplication

The product of two complex numbers z_1 and z_2 is found by multiplying them out in full and remembering that $i^2 = -1$.

$$z_1 z_2 = (x_1 + y_1 i)(x_2 + y_2 i) \quad (12)$$

$$= x_1 x_2 + x_1 y_2 i + y_1 x_2 i + y_1 y_2 i^2 \quad (13)$$

$$= (x_1 x_2 - y_1 y_2) + (x_1 y_2 + y_1 x_2) i \quad (14)$$



Example 3

Question

Multiply the complex numbers $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$.

Solution

By direct multiplication , we find

$$z_1 z_2 = (3+2i)(-1-4i) = -3-2i-12i-8i^2 = 5-14i$$



Cont...

The multiplication of complex numbers is both commutative and associate. i.e

$$z_1 z_2 = z_2 z_1 \quad (15)$$

$$(z_1 z_2) z_3 = z_1 (z_2 z_3) \quad (16)$$

$$|z_1 z_2| = |z_1| |z_2| \quad (17)$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2. \quad (18)$$



Example 4

Question

Verify that $|z_1 z_2| = |z_1| |z_2|$, holds for the product of $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$

Cont...

Solution

$$|z_1 z_2| = |5 - 14i| = \sqrt{5^2 + (-14)^2} = \sqrt{221}$$

We also find

$$\begin{aligned} |z_1| &= \sqrt{3^2 + 2^2} = \sqrt{13} \\ |z_2| &= \sqrt{(-1)^2 + (-4)^2} = \sqrt{17} \end{aligned}$$

and hence

$$|z_1||z_2| = \sqrt{13}\sqrt{17} = \sqrt{221} = |z_1 z_2|$$



Complex Conjugate

If z has the convenient form $x + yi$ then the complex conjugate, denoted by $\bar{z}$ may be found simply by changing the sign of the imaginary part, i.e, if

$$z = x + yi$$

then

$$\bar{z} = x - yi$$



Example 5

Question

Find the complex conjugate of

$$z = a + 2i + 3ib$$

Solution:

$$\bar{z} = a - (2 + 3b)i$$



Example 6

Question

Find the complex conjugate of the complex number $z = w^{(3y+2ix)}$ where $w = x + 5i$.



Cont...

Solution

In this case , w itself contains real and imaginary components and so must be written out in full, i.e.

$$z = w^{(3y+2ix)} = (x + 5i)^{(3y+2ix)}$$

Now we can replace each i by $-i$ to obtain

$$\bar{z} = (x - 5i)^{(3y-2ix)}$$

It can be shown that the product $z\bar{z}$ is real as required.



Properties

The following properties of complex conjugate are easily proved and others maybe derived from them.

If $z = x + yi$

then

(a) $\bar{\bar{z}} = z$

(b) $z + \bar{z} = 2 \operatorname{Re} z = 2x$

(c) $z - \bar{z} = 2 \operatorname{Im} z = 2yi$



Cont...

$$(d) \frac{z}{\bar{z}} = \left(\frac{x^2 - y^2}{x^2 + y^2} \right) + i \left(\frac{2xy}{x^2 + y^2} \right)$$

$$(e) \overline{z_1 z_2} = \overline{z_2} \overline{z_1} \text{ for two complex numbers } z_1 \text{ and } z_2.$$

$$(f) \overline{z_1 + z_2} = \overline{z_1} + \overline{z_2} \text{ for two complex numbers } z_1 \text{ and } z_2.$$



Division

The division of two complex numbers z_1 and z_2 bears some similarity to their multiplication.

Writing the quotient in the component form we obtain

$$\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i} \quad (19)$$



Cont...

In order to separate the real and the imaginary components of the quotient, we multiply both numerator by the complex conjugate of the denominator.

By definition , this process will leave the denominator as a real quantity.



Cont...

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{(x_1 + y_1 i)(x_2 + y_2 i)}{(x_2 + 2i)(x_2 - y_2 i)} = \frac{(x_1 x_2 + y_1 y_2) + (x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \\ &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}\end{aligned}$$

Hence we have separated quotient into real and imaginary components as required.



Example 7

Question

Express z in the form $x + iy$, when $z = \frac{3 - 2i}{-1 + 4i}$

Cont...

Solution

Multiplying numerator and denominator by the complex conjugate of the denominator, we obtain

$$\begin{aligned} z &= \frac{(3 - 2i)(-1 - 4i)}{(-1 + 4i)(-1 - 4i)} = \frac{-11 - 10i}{17} \\ &= \frac{-11}{17} - \frac{10}{17}i \end{aligned}$$



Properties

$$(a) \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$(b) \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

Exercise 1

1. Given $z_1 = 2 + 4i$, $z_2 = 3 - i$ and $z_3 = -3i$
Determine:

(a) $z_1 + z_2$

(b) $z_1 - z_2$

(c) $z_1 - z_3$

2. Find

(a) $(1 - 8i)(1 + 8i)$

(b) $(4 + i)(3i)$

(c) $(2 - i)(5 + 2i)$



Cont...

3. Write each in standard form.

$$(a) \frac{3 - i}{2 + 7i}$$

$$(b) \frac{9 - i}{8i}$$

$$(c) \frac{1 + 2i}{1 + 2i}$$



Polar Representation Of Numbers

The polar representation of a complex number is easier to manipulate. The polar representation makes use of the complex exponential function which is defined by

$$e^z = \exp z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \quad (20)$$

Note that we may have from (20)

$$e^{z_1} e^{z_2} = e^{z_1 + z_2} \quad (21)$$



Cont...

From (20), it immediately follows that for $z = i\theta$,

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \dots \quad (22)$$

$$= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots + i\theta - i\frac{\theta^3}{3!} + i\frac{\theta^5}{5!} - \dots \quad (23)$$

which is a sum of series expansion of $\cos \theta$ and $\sin \theta$



Cont...

and hence we obtain

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (24)$$

called the **Euler's equations**

It also follows from (24)

$$e^{in\theta} = \cos n\theta + i \sin n\theta \quad (25)$$

for all n .

Cont...

From Euler's equation (24) we deduce that

$$re^{i\theta} = r(\cos \theta + i \sin \theta) \quad (26)$$

$$= x + yi \quad (27)$$

Thus a complex number may be represented in the polar form

$$z = re^{i\theta} \quad (28)$$



Cont...

Now we can identify r with $|z|$ and θ with $\arg z$.

The angle θ lies conventionally in the range $-\pi < \theta \leq \pi$, but, since rotation by θ is the same as rotation by $2n\pi + \theta$, where n is any integer,

$$re^{i\theta} \equiv re^{i(\theta+2n\pi)} \quad (29)$$



Cont...

The algebra for the polar representation is different from that of the real and imaginary component representation, though, of course, the results are identical.

Similarly from the Cartesian plane, the polar to Cartesian co-ordinate transformation equation can be written as follows



Cont...

$$x = r \cos \theta \quad y = r \sin \theta \quad (30)$$

where $r = |z|$, the modulus of z , and $\theta = \tan^{-1} \left(\frac{y}{x} \right)$, the argument of z . Thus, a complex number in polar form is given by

$$z = x + iy \quad (31)$$

$$= r \cos \theta + ir \sin \theta \quad (32)$$

$$= r(\cos \theta + i \sin \theta) \quad (33)$$



Example 8

Question

Express $z = 3 + 4i$ in polar form.

Solution

Since $|z| = 5$ and $\arg(z) = 53.13^\circ$, hence we may write the polar form of z explicitly as

$$z = 5(\cos 53.13^\circ + i \sin 53.13^\circ)$$



MULTIPLICATION AND DIVISION IN POLAR FORM

Multiplication and division in polar form are particularly simple. The product of $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ is given by

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} \quad (34)$$

$$= r_1 r_2 e^{(\theta_1 + \theta_2)} \quad (35)$$

$$= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \quad (36)$$



Cont...

Division is equally simple in polar form; the quotient of z_1 and z_2 is given by

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \quad (37)$$

$$= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \quad (38)$$



De MOIVRE'S THEOREM

We now derive an extremely important theorem.

Since

$$(e^{i\theta})^n = e^{in\theta} \quad (39)$$

we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad (40)$$

where the identity

$$e^{in\theta} = \cos n\theta + i \sin n\theta \quad (41)$$

follows from the series definition of $e^{in\theta}$.



Cont...

The result is called **De Moivre's theorem**. The theorem is valid for all n whether real imaginary or complex.

The theorem will be applied to

- proves of trigonometric identities,
- finding the n th roots of unity, and
- solving polynomial equations with complex roots.



TRIGONOMETRIC IDENTITIES

The use of **De Moivre's theorem** in finding trigonometric identities is best illustrated by examples. We consider the expression of a multiple - angle function in terms of a polynomial in the single-angle function, and its converse.



Example 9

Question

Express $\sin 3\theta$ and $\cos 3\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.

Cont...

Solution

Using de Moivre's theorem

$$\begin{aligned}\cos 3\theta + i \sin 3\theta &= (\cos \theta + i \sin \theta)^3 \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \sin \theta \cos^2 \theta - \sin^3 \theta)\end{aligned}$$



Cont...

Solution

We can equate the real and imaginary part coefficients separately, i.e,

$$\begin{aligned}\cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$



Cont...

Solution

$$\begin{aligned}\sin 3\theta &= 3 \sin \theta \cos^2 \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta\end{aligned}$$

This method can clearly be applied to finding power expansions of $\cos n\theta$ for any positive integer n .



NOTE

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad (42)$$

$$z^n - \frac{1}{z^n} = 2i \sin n\theta \quad (43)$$

These equalities follow from simple applications of de Moivre's theorem

Cont...

i.e.

$$\begin{aligned} z_n + \frac{1}{z_n} &= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \\ &= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned}$$



Cont...

and

$$\begin{aligned} z^n - \frac{1}{z^n} &= (\cos \theta + i \sin \theta)^n - (\cos \theta + i \sin \theta)^{-n} \\ &= \cos n\theta + i \sin n\theta - \cos(-n\theta) + i \sin(n\theta) \\ &= \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta \\ &= 2i \sin n\theta \end{aligned}$$



Cont...

In particular case where $n = 1$

We have,

$$z + \frac{1}{z} = e^{i\theta} + e^{-i\theta} = 2 \cos \theta \quad (44)$$

$$z - \frac{1}{z} = e^{i\theta} - e^{-i\theta} = 2i \sin \theta \quad (45)$$



Example 10

Question

Find an expression for $\cos^3 \theta$ in terms of $\cos 3\theta$ and $\cos \theta$

Solution

$$\begin{aligned}\cos^3 \theta &= \frac{1}{2^3} \left(z + \frac{1}{z} \right)^3 \\ &= \frac{1}{8} \left(z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \right)\end{aligned}$$



Cont...

Solution

$$= \frac{1}{8}\left(z^3 + \frac{1}{z^3}\right) + \frac{3}{8}\left(z + \frac{1}{z}\right)$$

we find that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta$$

Exercise 2

Using De Moivre's theorem

1. Expand

(a) $\cos 4\theta$ and $\sin 4\theta$

(b) $\cos^3 \theta$ and $\sin^3 \theta$

2. Show that

(a) $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$

(b) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

(c) $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$



Finding The Nth Roots Of Unity

The equation $z^2 = 1$ has the familiar solutions $z = \pm 1$.

In complex numbers, we can solve the general equation $z^n = 1$. And we know that the equation has n solutions. In order to proceed, we rewrite the equation as

$$z^n = e^{2ik\pi} \quad (46)$$

where k is any integer.



Cont...

Now taking the n th root of each side of the equation, we find

$$z = e^{\frac{2ik\pi}{n}} \quad (47)$$

Hence, the solutions of $z^n = 1$ are

$$z_{1,2,\dots,n} = 1, e^{\frac{2i\pi}{n}}, \dots, e^{\frac{2i(n-1)\pi}{n}}, \quad (48)$$

corresponding to the values $k = 0, 1, 2, \dots, n - 1$



NOTE

Larger integer values of k do not give new solutions, since the roots already listed are simply cyclically repeated for $k = n, n + 1, n + 2$, etc.

Example 11

Question

Find the solutions of the equation $z^3 = 1$.

Solution

By applying the above method we find

$$z = e^{\frac{2ik\pi}{3}}$$

Hence the three solutions are

$$z_1 = e^{0i} = 1, z_2 = e^{\frac{2i\pi}{3}}, z_3 = e^{\frac{4i\pi}{3}}.$$



Cont...

Solution

We note that, as expected, the next solution for which $k = 3$, gives $z_4 = e^{\frac{6i\pi}{3}} = 1 = z_1$, so that there are only three separate solutions.



Solving Polynomial Equations

A third application of de Moivre's theorem is the **solution of polynomial equations**. Complex equations in the form of a polynomial relationship must first be solved for z in a similar fashion to the method for finding the n roots of real polynomial equations. Then the complex roots of z may be found.



Example 12

Question

Solve the equation

$$z^6 - z^5 + 4z^4 - 6z^3 + 2z^2 - 8z + 8 = 0$$

Solution

We first factorize to give

$$(z^3 - 2)(z^2 + 4)(z - 1) = 0$$

Hence $z^3 = 2$ or $z^2 = -4$ or $z = 1$. The solutions to the quadratic equation are $z = \pm 2i$;

Cont...

Solution

to find the complex cube roots, we first write the equation in the form

$$z^3 = 2 = 2e^{2ik\pi}$$

where k is any integer. If we now take the cube root, we get

$$z = 2^{\frac{1}{3}} e^{\frac{2ik\pi}{3}}$$



Cont...

Solution

To avoid the duplication of solutions, we use the fact that $-\pi < \arg z \leq \pi$ and find

$$\begin{aligned}z_1 &= 2^{\frac{1}{3}} \\z_2 &= 2^{\frac{1}{3}} e^{\frac{2\pi i}{3}} = 2^{\frac{1}{3}} \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \\z_3 &= 2^{\frac{1}{3}} e^{-\frac{2\pi i}{3}} = 2^{\frac{1}{3}} \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)\end{aligned}$$



Complex Logarithm And Complex Powers

We can define the logarithm of a complex number and we can use complex numbers as exponents. Let us denote the natural logarithm of a complex number z by

$$\omega = \ln z,$$

where the notation $\ln$ will be explained shortly. Thus, ω must satisfy

$$z = e^{\omega} \tag{49}$$



Cont...

we see that

$$z_1 z_2 = e^{\omega_1} e^{\omega_2} = e^{\omega_1 + \omega_2}$$

and taking logarithms of both sides we find

$$\ln(z_1 z_2) = \omega_1 + \omega_2 = \ln z_1 + \ln z_2 \quad (50)$$



Cont...

We may investigate further the properties of $\ln z$.
We have already noted that the argument of a complex number is multivalued , i.e.

$$\arg z = \theta + 2n\pi$$

where n is any integer. Thus, in polar form, the complex number z should strictly be written as

$$z = re^{i(\theta+2n\pi)} \quad (51)$$



Cont...

Taking the logarithm of both sides and using (28), we find

$$\ln z = \ln r + i(\theta + 2n\pi) \quad (52)$$

where $\ln r$ is the natural logarithm of the real positive quantity r and so is written normally. Thus from (52) we see that $\ln z$ is itself multivalued.

Cont...

To avoid this multivalued behaviour it is conventional to define another function $\ln z$, the principal value of $\ln z$, which is obtained from $\ln z$ by restricting the argument of z to lie in the range $-\pi \leq \theta \leq \pi$.

Example 13

Question

Evaluate $\ln(-i)$

Solution

By rewriting i as a complex exponential , we find

$$\ln(-i) = \ln \left[e^{i(-\pi/2 + 2n\pi)} \right] = i(-\pi/2 + 2n\pi) \quad (53)$$

where n is any integer.



Cont...

Solution

Hence

$$\ln(-i) = -\frac{\pi}{2}i, \frac{3\pi}{2}i \quad (54)$$

We note that, the principal value of $\ln(-i)$, is given by

$$\ln(-i) = -\frac{\pi}{2}i \quad (55)$$



Complex Exponential

Example

Simplify the expression $z = i^{-2i}$

Solution

First we take the logarithm of both sides of the equation:

$$\ln z = -2i \ln i \quad (56)$$



Cont...

Solution

Now inverting the process we find

$$e^{lnz} = z = e^{-2ilni} \quad (57)$$

We can write $i = e^{i(-\pi/2+2n\pi)}$, where n is any integer, and hence

$$\begin{aligned} lni &= \ln \left[e^{i(\pi/2+2n\pi)} \right] \\ &= i(\pi/2 + 2n\pi) \end{aligned}$$



Cont...

Solution

We can now simplify z to give

$$\begin{aligned}j^{-2i} &= e^{-2i \times i(\frac{\pi}{2} + 2n\pi)} \\ &= e^{\pi + 4n\pi}\end{aligned}$$

which is a real quantity rather than a complex one.

Exercise 3

Evaluate

1. $\ln(2 - 3i)$

2. $\ln(-2 + i)$

3. 2^{-1-i}

4. $(2 - i)^i$

Vector Algebra

Definition

The term **vector** is used by scientists to indicate a quantity that has both magnitude and direction.

Examples: displacement or velocity or force.

A vector is represented by an arrow or a directed line segment ($\longrightarrow$). The length of the arrow represents **magnitude** of the vector and the arrow points in the **direction of the vector**.

Cont...

We denote a vector by the letter **u**.

A vector $\overrightarrow{AB}$ has initial point **A** (the tail) and terminal point **B** (the tip). We say that **u** and **v** are equivalent (or equal) and we write **u = v**. The zero vector denoted by **0**, has length 0. **It is the only vector with no specific direction.**



Combining Vectors

Suppose a particle moves from A to B , so its displacement vector is $\overrightarrow{AB}$. The particle changes direction and moves from B to C , with displacement vector $\overrightarrow{BC}$.

The combined effect of these displacements is that the particle has moved from A to C .



Cont...

The resulting displacement vector $\overrightarrow{AC}$ is called the sum of $\overrightarrow{AB}$ and $\overrightarrow{BC}$ written as;

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

In general , if we start with vectors $\mathbf{u}$ and $\mathbf{v}$, we first move $\mathbf{v}$ so that its tail coincides with the tip of $\mathbf{u}$ and define the sum of $\mathbf{u}$ and $\mathbf{v}$ as follows.



Definition Of Vector Addition

1. If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$,
2. then the sum $\mathbf{u} + \mathbf{v}$ is the vector from the initial point of $\mathbf{u}$ to the final point $\mathbf{v}$.
3. If we place $\mathbf{u}$ and $\mathbf{v}$ so they start at the same point, then $\mathbf{u} + \mathbf{v}$ lies along the diagonal of the parallelogram with $\mathbf{u}$ and $\mathbf{v}$ as sides.
(Parallelogram Law.)



Scalar Multiplication

4. If c is a scalar and $\mathbf{u}$ is a vector then the scalar multiple $c\mathbf{u}$ is the vector
- (a) whose length is $|c|$ times the length of $\mathbf{u}$ and
 - (b) whose direction is the same as $\mathbf{u}$ if $c > 0$ and
 - (c) whose direction is opposite to $\mathbf{u}$ if $c < 0$.



Cont...

5. If $c = 0$ or $\mathbf{u} = 0$, then $c\mathbf{u} = 0$.
6. Two non-zero vectors are parallel if they are scalar multiples of one another.
7. The vector $-\mathbf{u} = (-1)\mathbf{u}$ has the same length as $\mathbf{u}$ but points in the opposite direction of $\mathbf{u}$. We call it the negative of $\mathbf{u}$. By the difference $\mathbf{u} - \mathbf{v}$ of two vectors we mean

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$



Parallel Vectors

Two vectors are **parallel** if they have the **same direction** or are in **exactly opposite directions**.

Suppose that a and b are parallel vectors. If they are parallel there must be a number c so that ,

$$a = cb$$

So, two vectors are parallel if one is a scalar multiple of the other.

Components Of A Vector

If we place the initial point of a vector a at the origin of a rectangular coordinates system, then the terminal point of a has coordinates of the form (a_1, a_2) or (a_1, a_2, a_3) , depending on whether our coordinate system is two or three -dimensional. These coordinates are called the **components of the vector** and we write

$$a = \langle a_1, a_2 \rangle \text{ or } a = \langle a_1, a_2, a_3 \rangle$$



Cont...

We use the notation $a = \langle a_1, a_2 \rangle$ for the ordered pair that refers to a vector so as not to confuse it with the ordered pair (a_1, a_2) that refers to a point in the plane.

In three dimensions, the vector $a = \overrightarrow{OP} = \langle a_1, a_2, a_3 \rangle$ is the position vector of the point $P(a_1, a_2, a_3)$.

Let's consider any other representation $\overrightarrow{AB}$ of a , where the initial point is $A(x_1, y_1, z_1)$ and the terminal point is $B(x_2, y_2, z_2)$.



Cont...

Then we must have

$$x_1 + a_1 = x_2, y_1 + a_2 = y_2 \text{ and } z_1 + a_3 = z_3$$

and so

$$a_1 = x_2 - x_1, a_2 = y_2 - y_1, \text{ and } a_3 = z_2 - z_1$$

Given the points $A(x_1, x_2, x_3)$ and $B(x_2, y_2, z_2)$, the vector with representation $\overrightarrow{AB}$ is

$$\overrightarrow{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$



Example 14

Question

Find the vector represented by the directed line segment with initial point $A(1, -3, 4)$ and terminal point $B(-2, 1, -1)$.

Solution

the vector corresponding to $\overrightarrow{AB}$ is

$$a = \langle -2 - 1, 1 - (-3), -1 - 4 \rangle = \langle -3, 4, -5 \rangle$$



Cont...

Magnitude

The magnitude or length of the vector is the length of any of its representations and is denoted by the symbol $|v|$. By using the distance formula to compute the length of a segment OP , we obtain the following formulas.

Cont...

The length of the two-dimensional vector $a = \langle a_1, a_2 \rangle$ is

$$|a| = \sqrt{a_1^2 + a_2^2}$$

The length of the three-dimensional vector $\langle a_1, a_2, a_3 \rangle$ is

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$



Cont...

If $a = \langle a_1, a_2 \rangle$ and $b = \langle b_1, b_2 \rangle$, then

$$a + b = \langle a_1 + b_1, a_2 + b_2 \rangle$$

$$a - b = \langle a_1 - b_1, a_2 - b_2 \rangle$$

$$ca = \langle ca_1, ca_2 \rangle$$

Similarly, for three-dimensional vectors,

Cont...

$$\langle a_1, a_2, a_3 \rangle + \langle b_1, b_2, b_3 \rangle = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\langle a_1, a_2, a_3 \rangle - \langle b_1, b_2, b_3 \rangle = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

$$c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$



Properties of Vectors

If a, b and c are vectors in V_n and k and d are scalars, then

$$(a) \quad a + b = b + a$$

$$(e) \quad a + 0 = a$$

$$(b) \quad a + (b + c) = (a + b) + c$$

$$(f) \quad (k + d)a = ka + da$$

$$(c) \quad a + (-a) = 0$$

$$(g) \quad (kd)a = k(da)$$

$$(d) \quad k(a + b) = ka + kb$$

$$(h) \quad 1a = a$$



The Standard Basis Vectors i, j and k

Three vectors in V_3 play a special role. Let

$$i = \langle 1, 0, 0 \rangle \quad j = \langle 0, 1, 0 \rangle \quad k = \langle 0, 0, 1 \rangle$$

These vectors i, j and k are called the **standard basis vectors**. They have **length 1** and **point in the directions** of the positive x, y and z — *axes*.

Similarly, in two dimensions we define $i = \langle 1, 0 \rangle$ and $j = \langle 0, 1 \rangle$



Cont...

$$\text{if } a = \langle a_1, a_2, a_3 \rangle = \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle$$

$$= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle = a_1 i + a_2 j + a_3 k$$

Thus any vector in V_3 can be expressed in terms of i, j, k and for instance,

$$\langle 1, -2, 6 \rangle = i - 2j + 6k$$

Similarly, in two dimensions, we can write

$$a = \langle a_1, a_2 \rangle = a_1 i + a_2 j$$



Example 15

Question

If $a = i + 2j - 3k$ and $b = 4i + 7k$, express the vector $2a + 3b$ in terms of i, j, k .

Solution

Using Properties 1,2,5,6, and 7 of vectors, we have

$$\begin{aligned} 2a + 3b &= 2(i + 2j - 3k) + 3(4i + 7k) \\ &= 2i + 4j - 6k + 12i + 21k = 14i + 4j + 15k \end{aligned}$$



Unit Vector

A **unit vector** is a vector whose **length** is **1**. For instance, $\hat{i}, \hat{j}, \hat{k}$ and are all unit vectors. In general, if $a \neq 0$, then the unit vector that has the same direction as a is

$$u = \frac{1}{|a|} a = \frac{a}{|a|} \quad (58)$$



Cont...

In order to verify this , we let $c = \frac{1}{|a|}$. Then $u = ca$ and c is a positive scalar, so u has the same direction as a . Also

$$|u| = |ca| = |c||a| = \frac{1}{|a|}|a| = 1 \quad (59)$$



Example 16

Question

Find the unit vector in the direction of the vector $2i - j - 2k$.

Solution

The given vector has length

$$|2i - j - 2k| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{9} = 3$$

So, the unit vector with the same direction is

$$\frac{1}{3}(2i - j - 2k) = \frac{2}{3}i - \frac{1}{3}j - \frac{2}{3}k$$



Dot Product

DEFINITION

If $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, then the dot product of a and b is the number $a \cdot b$ given by

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

It is a real number, that is, a scalar. For this reason, the **dot product** is sometimes called the **scalar product (or inner product)**.



Cont...

The dot product of two-dimensional vectors is defined in a similar fashion.

$$\langle a_1, a_2 \rangle \cdot \langle b_1, b_2 \rangle = a_1 b_1 + a_2 b_2$$

Example 17

1. $\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle = 2(3) + 4(-1) = 2$

2. $\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -\frac{1}{2} \rangle =$
 $(-1)(6) + 7(2) + 4(-\frac{1}{2}) = 6$

3. $(i + 2j - 3k) \cdot (2j - k) =$
 $1(0) + 2(2) + (-3)(-1) = 7$

Properties of The Dot Product

The dot product obeys many of the laws that hold for ordinary products of real numbers. These are stated in the following theorem.

Cont...

If a, b and c are vectors in V_3 and k is a scalar ,
then

- $a \cdot a = |a|^2$
- $a \cdot b = b \cdot a$
- $a \cdot (b + c) = a \cdot b + a \cdot c$
- $0 \cdot a = 0$
- $(ka) \cdot b = k(a \cdot b) = a \cdot (kb)$
- $a \cdot b = 0$, a and b are mutually perpendicular or $a = 0$ or $b = 0$.



Exercise 4

Proof the above properties as exercise.

Theorem 1

If θ is the angle between the vectors a and b , then

$$a \cdot b = |a||b|\cos\theta$$

Example 18

Question

If the vector a and b have length 4 and 6, and the angle between them is $\pi/3$, find $a \cdot b$

Solution

Using Theorem 1, we have

$$a \cdot b = |a||b| \cos(\pi/3) = 4 \cdot 6 \cdot \frac{1}{2} = 12$$

Question

Given two vectors $\mathbf{a} = \langle 2, 2, 1 \rangle$ and $\mathbf{b} = \langle 1, 1, -2 \rangle$ in a three-dimensional space, find the angle between these two vectors.

Cont...

Solution

To find the angle between two vectors we have,

$$\cos \theta = \frac{a \cdot b}{|a||b|} = \frac{2}{3\sqrt{38}}$$

so the angle between a and b is

$$\theta = \cos^{-1} \left(\frac{2}{3\sqrt{38}} \right) = 74.2068^\circ$$



Cont...

Two non-zero vectors a and b are called **perpendicular or orthogonal** if the angle between them is $\theta = \pi/2$

$$a \cdot b = |a||b| \cos(\pi/2) = 0$$

and conversely if

$$a \cdot b = 0$$

the $\cos \theta = 0$, so $\theta = \pi/2$. The zero vector is considered to be perpendicular to all vectors.



Theorem 2

Two vectors a and b are orthogonal if and only if
 $a \cdot b = 0$

Example 19

Question

Show that $2i + 2j - k$ is perpendicular to $5i - 4j + 2k$

Solution

Since

$$\begin{aligned}(2i + 2j - k) \cdot (5i - 4j + 2k) &= \\ 2(5) + 2(-4) + (-1)(2) &= 0\end{aligned}$$

these vectors are perpendicular.



Projection

The scalar projection of b onto a (also called the component of b along a) is defined to be the signed magnitude of the vector projection, which is the number $|b| \cos \theta$, where θ is the angle between a and b . This is denoted by $\text{comp}_a b$. It's negative if $\pi/2 < \theta \leq \pi$.



Cont...

The equation

$$a \cdot b = |a||b| \cos \theta = |a|(|b| \cos \theta)$$

shows that the dot product of a and b can be interpreted as the length of a times the scalar projection of b onto a . Also,

$$|b| \cos \theta = \frac{a \cdot b}{|a|} = \frac{a}{|a|} \cdot b$$

the component of b along a can be computed by taking the dot product of b with the unit vector in the direction of a .



Cont...

We summarize these ideas as follows.

1. Scalar projection of b onto a :

$$\text{comp}_a b = \frac{a \cdot b}{|a|}$$

2. Vector projection of b onto a :

$$\text{proj}_a b = \left(\frac{a \cdot b}{|a|} \right) \frac{a}{|a|} = \frac{a \cdot b}{|a|^2} a$$

The vector projection is the scalar projection times the unit vector in the direction of a .



Example 20

Question

Find the scalar projection and vector projection of $b = \langle 1, 1, 2 \rangle$ onto $a = \langle -2, 3, 1 \rangle$.

Solution

Since $|a| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{14}$, the scalar projection of b onto a is

$$\text{comp}_a b = \frac{a \cdot b}{|a|} = \frac{(-2)(1) + 3(1) + 1(2)}{\sqrt{14}} = \frac{3}{\sqrt{14}}$$



Cont...

Solution

The vector projection is this scalar projection times the unit vector in the direction of a :

$$\text{proj}_a b = \frac{3}{\sqrt{14}} \frac{a}{|a|} = \frac{3}{14} a = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14} \right\rangle$$



The Cross Product

Given two nonzero vectors $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, it is very useful to be able to find a nonzero vector that is **perpendicular** to both a and b .

If $c = \langle c_1, c_2, c_3 \rangle$ is such a vector, then $a \cdot c = 0$ and $b \cdot c = 0$ and so

$$a_1 c_1 + a_2 c_2 + a_3 c_3 = 0 \quad (60)$$



Cont...

$$b_1c_1 + b_2c_2 + b_3c_3 = 0 \quad (61)$$

To eliminate c_3 we multiply (60) by b_3 and (61) by a_3 and subtract:

$$(a_1b_3 - a_3b_1)c_1 + (a_2b_3 - a_3b_2)c_2 = 0 \quad (62)$$



Cont...

Equation (62) has the form $pc_1 + qc_2 = 0$, for which an obvious solution is

$c_1 = q$ and $c_2 = -p$.

So a solution of (60) is

$$c_1 = a_2b_3 - a_3b_2, c_2 = a_3b_1 - a_1b_3 \quad (63)$$

Substituting these values into (60) and (61), we then get

$$c_3 = a_1b_2 - a_2b_1$$



Cont...

This means that a vector perpendicular to both a and b is

$$\langle c_1, c_2, c_3 \rangle = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$$

The resulting vector is called the **cross product** of a and b and it's denoted by $a \times b$



Cont...

Definition

If $a = \langle a_1, a_2, a_3 \rangle$ and $b = \langle b_1, b_2, b_3 \rangle$, then the cross product of a and b is the vector

$$a \times b = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle \quad (64)$$

Cont...

The cross product $a \times b$ of two vectors a and b , unlike the dot product $a \cdot b$, is a **vector**. For this reason it is also called the **vector product**.

Note that $a \times b$ is defined only when a and b are three-dimensional vectors.